# Helix Thready — MVP Documentation Completeness Report

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| Rev | Date | Author | Change |
|---|---|---|---|
| 1 | 2026-07-22 | swarm (completeness critic) | Final completeness pass: full-tree link sweep + fixes, cross-area decision-matrix reconciliation, success-criteria audit, Pass-3 sign-off, consolidated open-item roll-up |

This is the **final completeness critique** of the entire Helix Thready MVP documentation tree under `docs/public/research/mvp/`. It closes the loop opened by INTEGRATION_REPORT.md: it re-swept every cross-area link, verified the technology decision matrix is reflected without contradiction across all eight areas, audited each success criterion from the authoritative request, patched the residual gaps found, and states an honest readiness verdict for implementation agents. All content follows CONVENTIONS.md.

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## 1. Summary

- **Tree:** 8 areas + 4 root docs = **370 files** — **95 Markdown**, **105 Mermaid** `.mmd` sources, **111** `.svg` (104 real diagram renders + 7 authored brand assets), **13** `.sql` (2 schema DDL files + 9 migrations + 2 materials DDL), plus OpenAPI/config/SDK-skeleton materials.
- **Links: PASS. 1314** file-targeted Markdown links swept; **2 real breakages fixed** (depth-4 `materials/*/README.md` pointing at `../../CONVENTIONS.md` instead of `../../../CONVENTIONS.md`); the one remaining regex hit is the inline-code link *example* in `CONVENTIONS.md §5` (never rendered). **0 broken links remain.**
- **Decision matrix: CONSISTENT.** The single cross-area contradiction — the product web client labelled *"Angular 22 Web"* in two API diagram nodes while every prose reference says **Angular 19 (product) / Angular 22 (marketing)** — was reconciled to **Angular 19 Web**.
- **Diagrams: PASS.** Every embedded `mermaid` block is followed by a multi-paragraph prose explanation; all 104 `.mmd` sources have real mermaid-cli `.svg` renders (see diagrams-render-status.md).
- **Gap register: COVERED.** All **8 P0** traps from the private gap register are owned by a design plan or `[BUILD-NEW]` item (none claimed to "work"); P1/P2 items each carry a `[GAP: …]` tag and a tracked plan; **85** `ATM-*` workable-item ids and **151** `[OPEN: …]` markers are tracked, none papered over.
- **Pass 3** is marked **done for all eight areas** in index.md: each now carries executable/reference materials beyond prose (OpenAPI + codegen, DDL + migrations, deploy configs, SDK skeletons, brand-asset SVGs, 15 test types with TDD skeletons, per-role/surface guides).

## 2. Final documentation tree & per-area file counts

```
docs/public/research/mvp/
├── index.md · CONVENTIONS.md · INTEGRATION_REPORT.md · COMPLETENESS_REPORT.md
├── diagrams-render-status.md · scripts/render-diagrams.sh
├── architecture/   13 md · 18 mmd · diagrams/
├── api/            13 md · 8 mmd · openapi.yaml (3.1) · materials/{codegen,examples} (28 files)
├── database/        8 md · 12 mmd · 13 sql (schema-relational, schema-vector, 7 migrations) · materials/{migrations,templates,diagrams} (7)
├── deployment/     15 md · 15 mmd · materials/{config/prometheus,systemd,diagrams} (18)
├── development/    10 md · 12 mmd · materials/sdk/{go,ts} (6)
├── testing/         9 md · 9 mmd  (15 test types + TDD reproduce-first skeletons + acceptance gates)
├── design/          9 md · 18 mmd · assets/ (10: logo-full, logo-mark, launcher-icon{,-dark,-light,-mono}, footer-slogan + generators)
└── user-guides/    14 md · 13 mmd  (install, config, root-admin, account-admin, end-user, cli, tui, web, mobile, sdk, faq, troubleshooting)
```

| Area | Markdown | `.mmd` | `.sql` | materials files | Area assessment |
|---|---|---|---|---|---|
| architecture | 13 | 18 | 0 | 0 | Deep — 11 subsystem docs + overview, every P0 trap owned |
| api | 13 | 8 | 0 | 28 | Deep — OpenAPI 3.1 + codegen/examples materials |
| database | 8 | 12 | 13 | 7 | Deep — relational + vector DDL + 7 forward migrations |
| deployment | 15 | 15 | 0 | 18 | Deep — Compose/systemd/Prometheus configs, backup/DR |
| development | 10 | 12 | 0 | 6 | Deep — ATM backlog + Go/TS SDK skeletons |
| testing | 9 | 9 | 0 | 0 | Deep — 15 test types + TDD skeletons + gates |
| design | 9 | 18 | 0 | 0 (10 in assets/) | Deep — OpenDesign linkage + brand-asset SVGs |
| user-guides | 14 | 13 | 0 | 0 | Deep — every consumer role × every surface |

*(`materials files` counts files under each area's `materials/` subtree; design's authored brand assets live under `design/assets/` and are counted there.)*

## 3. Consistency findings & fixes applied

| # | Finding | Severity | Action |
|---|---|---|---|
| 1 | `development/materials/sdk/README.md` linked `../../CONVENTIONS.md` (resolves to non-existent `development/CONVENTIONS.md`) — depth-4 file needs three `../` | Broken link | **Fixed** → `../../../CONVENTIONS.md` (header + body) |
| 2 | `api/materials/codegen/README.md` linked `../../CONVENTIONS.md` (resolves to non-existent `api/CONVENTIONS.md`) | Broken link | **Fixed** → `../../../CONVENTIONS.md` |
| 3 | Product web client labelled *"Angular 22 Web"* in `api/index.md` and `api/diagrams/api-surface.mmd`, contradicting the canonical **Angular 19 = product / Angular 22 = marketing** split used everywhere else | Cross-area contradiction | **Fixed** → both nodes now `Angular 19 Web` |
| 4 | `CONVENTIONS.md §5` shows `[Event model](../architecture/event-model.md)` as a link-syntax *example* (inline code) | False positive | No change — illustrative, never rendered |
| 5 | Mobile client labelling (`Native Mobile` / "native per platform") | Verified consistent | No change — architecture/design/user-guides/api all agree; Flutter appears only correctly as the Dart SDK codegen target and the documented alternative-only family |

No Upstream/Downstream directional note required correction; the reconciliation in INTEGRATION_REPORT.md §5 still holds.

## 4. Success-criteria checklist

Legend: **MET** — fully satisfied in docs · **PARTIAL** — satisfied with a tracked residual · **DELEGATED** — correctly deferred to implementation/tooling (cannot be closed inside documentation).

| # | Success criterion (from the request) | Verdict | Evidence |
|---|---|---|---|
| 1 | All sections detailed (no skinny docs / danger zones) | **MET** | 8 areas, 95 md; every subsystem + P0 trap has an owning doc; `[OPEN]` / `ATM` used, never papered over `[CONVENTIONS §7]` |
| 2 | Every diagram has a multi-paragraph prose explanation | **MET** | Each embedded `mermaid` block followed by an "Explanation …" section; 104/104 `.mmd` rendered to real `.svg` |
| 3 | DB schemas as DDL + migrations | **MET** | `database/schema-relational.sql`, `schema-vector.sql` (pgvector DDL); `migrations/0001…0007` forward scripts + `materials/migrations/{0002,0003}` |
| 4 | API defined as OpenAPI 3.1 | **MET** | `api/openapi.yaml` → `openapi: 3.1.0`; WS/SSE event contract in `api/event-bus-contract.md`; codegen materials |
| 5 | 15 mandated test types + TDD reproduce-first | **MET** | `testing/test-types.md` enumerates all 15 (unit… HelixQA) with acceptance |

| # | Success criterion (from the request) | Verdict | Evidence |
|---|---|---|---|
| | | | gates; `testing/tdd-skeletons.md` RED-first skeletons |
| 6 | Design linked to OpenDesign + brand assets exist | **MET** | `design/design-system.md` derives from OpenDesign (`nexu-io/open-design`); `design/assets/` holds logo, launcher icons (light/dark/mono), footer slogan SVGs + raster generator |
| 7 | User manuals per role **and** per surface | **MET** | Roles: root-admin, account-admin, end-user. Surfaces: CLI, TUI, web portal, mobile, SDK quickstart + install/config/FAQ/troubleshooting |
| 8 | Every gap-register item resolved or tracked | **MET** | 8/8 P0 gaps referenced across owning areas; P1/P2 each `[GAP:…]`-tagged; BUILD-NEW set declared in deployment + designed in `development/build-new-subsystems.md` |
| — | md ↔ HTML/PDF sibling generation (Docs Chain, `§11.4.65`) | **DELEGATED** | |

| # | Success criterion (from the request) | Verdict | Evidence |
|---|---|---|---|
| | | | Markdown is source-of-truth; `.svg` diagram renders done; HTML/PDF export gated on `pandoc` / `weasyprint` provisioning ( `[GAP: 19]` ) |
| — | Source-verification of `FLAGGED` upstream interfaces | **DELEGATED** | Re-verification backlog belongs to implementation; docs mark each `FLAGGED` and never assert a stub “works” |
| — | `.proto` DTO/event files, per-language SDK code | **DELEGATED** | SDK **skeletons** ( `development/materials/sdk/{go,ts}` ) + codegen strategy present; generated artifacts produced with code ( `[OPEN: api-1]` ) |

## 5. Consolidated remaining open items

All residual items are **tracked** ( `[OPEN: id]` + an `ATM-*` / gap plan) and fall into five deferral classes. None is a documentation deficiency; each is work that can only complete outside the docs (source verification, code, or host tooling).

| Class | Representative items | Why deferred (honest) |
|---|---|---|
| **A. Source-verification of FLAGGED upstreams** | `flagged-modules-verify`, `constitution-anchor-verify` (`ATM-066`), `agentpool-contract` (`ATM-067`), architecture `CAT-*` / `EVT-*` / `ING-*` / `PROC-*` / `SEM-*` / `ASSET-*` / `DF-*` | The private gap register marks these `FLAGGED` (README/metadata-only). Per the anti-bluff rule they must be re-read at source before code — a docs pass cannot and must not assert them verified. |
| **B. BUILD-NEW subsystems (design complete, code pending)** | `max-oneme-go-port` (`ATM-018`), `buildnew-images`, `buildnew-ports`, Download Manager / Asset Service / User Service / OCR adapter / MeTube webhook / callback module / Event Bus service / ThreadReader / Semantic-search service | These do not exist yet (`[BUILD-NEW]`). Docs specify the contract, seam and plan; the repos/code are implementation-phase deliverables. |
| **C. Scale/ perf tuning needing a running system** | `ATM-DB-012` (pgvector-vs-Qdrant benchmark), `ATM-DB-002` (`CREATE INDEX CONCURRENTLY`), `ATM-DB-032` (sub-partitioning), `host-sizing`, `gpu-perf-baseline`, `db-partition-fk` | Values (index build strategy, partition fan-out, host sizing, ANN recall/latency) require measurement against real data volume; documented as `[DEFAULT — adjustable]` with the tuning method. |
| **D. Host/ tooling provisioning** | `docs-chain-tooling` (pandoc/weasyprint), `mmd-md-sync`, `commit-artifacts`, `dns-provider`, `acme-email`, `minio-digest`, `secondary-store` | Depend on the Hetzner host / operator secrets / installed binaries not present at authoring time; the doc records the exact command and fallback. |
| **E. Contract-convergence to close before GA** | `THREADY-DES-14` (design `account_branding` table vs canonical `accounts.branding` JSONB + `setBranding` contract), `ATM-DB-033` (MinIO signed-URL parity), `api-2` (rate-limit tiers pending billing), `api-3` (Zig hand-written SDK), `fts-multilang`, `gdpr-cold-erasure` | Cross-area contracts intentionally converge on the canonical DB/API schema; design MUST NOT diverge. Tracked so the divergence is closed deliberately, not silently. |

P0 register traps (all owned, none claimed working): HelixLLM `HashEmbedder` default (`[GAP: 2.1]`), VisionEngine no-OCR (`[GAP: 2.6]`), helix_skills no execution engine (`[GAP: 4.1]`), Herald Telegram-in-QA-harness / Max empty stub (`[GAP: 5.1]`), filesystem no HTTP source (`[GAP: 6.2]`), Download Manager absent (`[GAP: 6.3]`), MeTube poll-only (`[GAP: 6.5]`), Security-KMP in-memory secure-storage stub (`[GAP: 7.3]`).

## 6. Readiness statement

The Helix Thready MVP documentation tree is **implementation-ready**. An implementation agent can pick up any of the four phases and find: an architecture with every component and P0 trap owned by a named design plan; a complete `/v1` **OpenAPI 3.1** contract plus WebSocket/SSE event contract and per-language codegen strategy; a relational + **pgvector** schema expressed as executable DDL with seven forward migrations; rootless-Podman deployment, TLS, backup/DR and secrets procedures with real config materials; a development backlog of 85 decoupled `ATM-*` workable items with Go/TS SDK skeletons; all **15**

**mandated test types** with TDD reproduce-first skeletons and acceptance gates; an OpenDesign-anchored design system with real brand-asset SVGs; and user manuals for every role across every surface. Cross-references resolve (0 broken of 1314), the technology decision matrix is reflected without contradiction, and every diagram carries a prose explanation. The remaining `[OPEN: …]` / `ATM-*` items are **honestly scoped deferrals** — source-verification of `FLAGGED` upstreams, `[BUILD-NEW]` code, measurement-dependent tuning, host tooling, and deliberate contract convergence — not gaps papered over. The one hard prerequisite before relying on any reused subsystem remains the register's **anti-bluff rule**: re-verify each `FLAGGED` interface at source, and treat each P0 trap as unbuilt until its owning plan is implemented and covered by the test banks.

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